Yixian Chen, Ph.D.

Department of Geography and the Environment, The University of Alabama, Tuscaloosa, AL 35401, USA
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RESEARCH INTERESTS

Physical Hydrology and Flood Inundation Mapping | River Systems and Water Resources | Geomorphology (Gully and Fluvial Processes) | Soil Erosion, Land Use, and Soil and Water Conservation | Geohazards: Floods, Extreme-Rainstorm Erosion, and Slope Failure | Remote Sensing, UAV, and LiDAR-Based Field Measurement | Geospatial Machine Learning

EDUCATION

2014.09 - 2021.01
Ph.D. in *Soil Science*
University of Chinese Academy of Sciences, Yangling & Beijing, China
Advisor: *Juying Jiao* (ResearchGate)
Dissertation: *Morphological Spatial Variation and Development Mechanisms of Gullies on the Chinese Loess Plateau*

2019.10 - 2020.10
Visiting Ph.D. Student
University of Liège, Liège, Belgium
Advisor: *Matthias Vanmaercke* (Google Scholar)
Research: *Sub-Continental Gully Density Prediction Based on a Machine Learning Approach*

2010.09 - 2014.06
B.S. in *Soil and Water Conservation and Desertification Control*
Guizhou University, Guiyang, China

PROFESSIONAL EXPERIENCE

2026.04 - Present
Research Scientist
The University of Alabama, Tuscaloosa, USA
Advisor: *Sagy Cohen* (Lab website)
Research: *Optimal Synthetic Rating Curve Calibration for the U.S. Operational Flood Inundation Mapping using Large-Scale Benchmark Flood Maps*

2024.05 - 2026.03
Postdoctoral Fellow
The University of Alabama, Tuscaloosa, USA
Advisor: *Sagy Cohen*
Research: *Improving Spatial and Temporal River Slope Data for the U.S. Operational Flood Inundation Mapping*

2021.03 - 2024.04
Postdoctoral Researcher
Northwest University, Xi'an, China
Advisor: *Ninglian Wang* (ResearchGate)
Research: *Morphological Characteristics and Volume Estimating Models of Gullies on the Chinese Loess Plateau*

Concurrently Postdoctoral Researcher, Katholieke Universiteit Leuven (KU Leuven), Leuven, Belgium, 2022.07 - 2023.06 | Advisor: *Matthias Vanmaercke* | Research: *Data-Driven Prediction of Gully Densities at the Global Scale*

RESEARCH GRANTS

Selecting the Optimal Synthetic Rating Curve Calibration Approach for Operational Flood Inundation Mapping using Large-Scale Benchmark Flood Maps
NOAA CIROH FY26 Research Project, 2026.07-2027.03, Total $300K USD. Role: PI

The Affecting Mechanisms of Valley Bank Gully on Erosion of Downslope Bank Slope in the Hilly-Gully Region of the Loess Plateau, China
National Natural Science Foundation of China (NSFC) Young Scientist Project, 2025-2027, ~$41K USD. Role: PI

Morphological Characteristics and Volume Estimating Models of Gullies on the Chinese Loess Plateau
China Postdoctoral General Project, 2021-2023, ~$11K USD. Role: PI

Morphological Characteristics and Volume Estimating Models of Gullies on the SW Chinese Loess Plateau
Natural Science Basic Research Program of Shaanxi Province, China, 2022-2023, ~$6.9K USD. Role: PI

Assessing the Global Hydrological Footprint of Gullies on Soil Moisture and Flood Risks
Submitted Senior Postdoctoral Fellowship Proposal to the Research Foundation - Flanders (FWO), November 2025, Not Selected. Role: PI

Quantifying Gully Erosion and its Impacts on Catchment Sediment Yields at the Global Scale
Submitted Postdoctoral Fellowship Proposal to the National Fund for Scientific Research of Belgium, 2021, Not Selected. Role: PI

PUBLICATIONS (FULL LIST AVAILABLE ON GOOGLE SCHOLAR)

Author of **24** articles in SCI journals (5 as first author; 1 as second author, with my supervisor as first and corresponding author), and **10** articles in Chinese journals (1 as first author and 1 as second author). As of October 2026: 819 citations, h-index 18, i10-index 26.

In review & preparation:

1. **Chen, Y.**, Cohen, S., Dhital, S., Devi, D., & Baruah, A. (2025). Spatial and Temporal Variability in Global River Slopes from the Surface Water and Ocean Topography (SWOT) Satellite. *Geophysical Research Letters*. Under revision. [link]
2. **Chen, Y.**, Dhital, S., Baruah, A., Devi, D., Munasinghe, D., & Cohen, S. RiverJoin: A Framework for Spatially Joining River Flowlines across Heterogeneous Hydrographic Datasets to Enable Bidirectional Attribute Transfer. In preparation.
3. Li, J., Jiao, J.*, Chen, Y., Ren, X., Zhang, Z., **Chen, Y.**, Yang, B., Xu, Q., Liu, L., & Jian, J. Quantifying Spatial Differentiation and Environmental Controls of Basic Gully Morphological Parameters on the Tibetan Plateau. *Ecological Indicators*. Under review.

Peer Reviewed: (* denotes corresponding author)

1. **Chen, Y.**, De Geeter, S., Poesen, J., Matthews, F., Campforts, B., Borrelli, P., ... & Vanmaercke, M. (2026). Global Patterns of Gully Occurrence and Their Sensitivity to Environmental Changes. *International Soil and Water Conservation Research*, 14, 100572. [link]
2. **Chen, Y.**, Cohen, S., Baruah, A., Devi, D., Dhital, S., Tian, D., & Munasinghe, D. (2025). Improved River Slope Datasets for the United States Operational Flood Inundation Mapping Hydrofabric and Next Generation Water Resources Modeling Hydrofabric. *Scientific Data*, 12(1), 1657. [link]
3. **Chen, Y.**, Jiao, J., Yan, X., Li, J., Vanmaercke, M., & Wang, N. (2024). Response of Gully Morphology and Density to the Spatial and Rainy-Season Monthly Variation of Rainfall at the Regional Scale of the Chinese Loess Plateau. *Catena*, 236, 107773. [link]
4. **Chen, Y.**, Vanmaercke, M., Jiao, J., Bai, L., Tang, B., Wang, N., et al. (2023). Quantifying the Importance of Different Erosion Processes and Soil and Water Conservation Measure Collapses Following an Extreme Rainstorm in the Chinese Loess Plateau. *Land Degradation & Development*, 34(2), 403-422. [link]
5. Vanmaercke, M.*, **Chen, Y.**, Haregeweyn, N., De Geeter, S., Campforts, B., Heyndrickx, W., et al. (2020). Predicting Gully Densities at Sub-Continental Scales: A Case Study for the Horn of Africa. *Earth Surface Processes and Landforms*, 45(15), 3763-3779. [link]
6. **Chen, Y.**, Jiao, J., Wei, Y., Zhao, H., Yu, W., Cao, B., et al. (2019). Accuracy Assessment of the Planar Morphology of Valley Bank Gullies Extracted with High Resolution Remote Sensing Imagery on the Loess Plateau, China. *International Journal of Environmental Research and Public Health*, 16(3), 369. [link]
7. **Chen, Y.**, Jiao, J., Wei, Y., & Zhao, H. (2017). Characteristics of Gully Development in Northern Shaanxi Hilly Loess Region Before Grain-for-Green Programme (1976-1997). *Transactions of the Chinese Society of Agricultural Engineering*, 33(17), 120-127. (EI-indexed, in Chinese with English abstract) [link]
8. Li, J., Jiao, J., **Chen, Y.**, Chen, Y., Ren, X., Zhang, Z., et al. (2026). Precipitation Buffers Gully Erosion Through Vegetation Cover Across the Tibetan Plateau. *Journal of Environmental Management*, 419, 131173. [link]
9. Baruah, A., Munasinghe, D., Cohen, S., Abdelkader, M., Devi, D., **Chen, Y.**, Vergara, H., & McDermott, R. (2026). Brief Communication: Rise of the Guadalupe River - A Multifaceted Post Event Analysis of 4 July 2025, Flood in Central Texas. *Natural Hazards and Earth System Sciences*, 26(10), 4753-4762. [link]
10. Devi, D., Dhital, S., Munasinghe, D., Cohen, S., Baruah, A., **Chen, Y.**, ... & Pruitt, C. (2026). A Framework for the Evaluation of Flood Inundation Predictions over Extensive Benchmark Databases. *Environmental Modelling & Software*, 196, 106786. [link]
11. Liu, H., Wang, C., **Chen, Y.**, Vanmaercke, M., Cruse, R. M., Long, Y., ... & Wang, E. (2026). Regional-Scale Cropland Gully Density Prediction Using Machine Learning Approach Enhanced by Unequal Probability Stratified Sampling. *Catena*, 262, 109700. [link]

12. Baruah, A., Dhital, S., Cohen, S., Tran, T. N. D., Elhaddad, H., Watts, C. L., Devi, D., **Chen, Y.**, & Pruitt, C. (2025). FIMserv v. 1.0: A Tool for Streamlining Flood Inundation Mapping (FIM) Using the United States Operational Hydrological Forecasting Framework. *Environmental Modelling & Software*, 192, 106581. [link]

13. Li, J., Jiao, J., **Chen, Y.**, Cao, X., Chen, T., Cheng, Y., et al. (2025). Elementary Study on Gully Classification on the Qinghai-Xizang Plateau. *Journal of Soil and Water Conservation*, (01), 1-9. (In Chinese with English abstract) [link]

14. Li, J., Chen, Y., Jiao, J., Cao, X., **Chen, Y.**, Chen, T., ... & Bai, L. (2024). Modeling Gully Initiation by Two Codeless Nonlinear Methods: A Case Study in a Small Watershed on the Tibetan Plateau. *International Soil and Water Conservation Research*, 12(4), 747-760. [link]

15. Jiang, Z., Hu, S., Deng, H., Wang, N., Zhang, F., Wang, L., Wu, S., Wang, X., Cao, Z., **Chen, Y.**, & Li, S. (2024). Detection and Automatic Identification of Loess Sinkholes from the Perspective of LiDAR Point Clouds and Deep Learning Algorithm. *Geomorphology*, 465, 109404. [link]

16. Li, J., Chen, Y., Jiao, J., **Chen, Y.**, Chen, T., Zhao, C., et al. (2024). Gully Erosion Susceptibility Maps and Influence Factor Analysis in the Lhasa River Basin on the Tibetan Plateau, Based on Machine Learning Algorithms. *Catena*, 235, 107695. [link]

17. Tang, B., Jiao, J., **Chen, Y.**, & Liang, Y. (2024). Rill Investigation on Landslide Scar Highlights the Shortcomings of Current Knowledge on Erosion of Shallow Landslides. *Land Degradation & Development*, 35(4), 1495-1510. [link]

18. Hou, J., Wang, N., Shi, C., **Chen, Y.**, Zhao, M., & Fang, Z. (2024). Study on Phenological Changes of Lake Ice in Mongolia Plateau from 2000 to 2021. *Journal of Glaciology & Geocryology*, (05), 1537-1550. (In Chinese with English abstract) [link]

19. Borrelli, P., Alewell, C., Yang, J. E., Bezak, N., **Chen, Y.**, Fenta, A. A., et al. (2023). Towards a Better Understanding of Pathways of Multiple Co-Occurring Erosion Processes on Global Cropland. *International Soil and Water Conservation Research*, 11(4), 713-725. [link]

20. Li, J., Wang, H., Chen, Y., **Chen, Y.**, Jiao, J., Chen, T., et al. (2023). Can CATPCA Be Utilized for Spatial Modeling? A Case of the Generation Susceptibility of Gully Head in a Watershed. *Catena*, 232, 107456. [link]

21. Li, J., Zhao, C., Chen, T., Jiao, J., Wang, H., Lin, H., Wang, N., **Chen, Y.**, et al. (2022). Gully Erosion on Alluvial Fans Can Be Mitigated by Altering the Hydrological Connectivity Between an Alluvial Fan and the Contributing Catchment: A Study in the Lhasa River Basin. *Land Degradation & Development*, 33(8), 1170-1183. [link]

22. Li, J., Chen, Y., Jiao, J., Chen, T., **Chen, Y.**, Zhao, W., et al. (2022). Detecting Gully Occurrence Risks Using Multivariate Nonlinear Spatial Modeling in the Lhasa River Basin of China. *Transactions of the Chinese Society of Agricultural Engineering*, 38(17), 73-82. (EI-indexed, in Chinese with English abstract) [link]

23. Chen, T., Jiao, J., **Chen, Y.**, Lin, H., Wang, H., & Bai, L. (2021). Distribution and Land Use Characteristics of Alluvial Fans in the Lhasa River Basin, Tibet. *Journal of Geographical Sciences*, 31(10), 1437-1452. [link]

24. Wang, N., Jiao, J., Bai, L., Zhang, Y., **Chen, Y.**, Tang, B., et al. (2020). Magnitude of Soil Erosion in Small Catchments with Different Land Use Patterns Under an Extreme Rainstorm Event over the Northern Loess Plateau, China. *Catena*, 195, 104780. [link]

25. Tang, B., Jiao, J., Zhang, Y., **Chen, Y.**, Wang, N., & Bai, L. (2020). The Magnitude of Soil Erosion on Hillslopes with Different Land Use Patterns Under an Extreme Rainstorm on the Northern Loess Plateau, China. *Soil and Tillage Research*, 204, 104716. [link]

26. Bai, L., Wang, N., Jiao, J., **Chen, Y.**, Tang, B., Wang, H., et al. (2020). Soil Erosion and Sediment Interception by Check Dams in a Watershed for an Extreme Rainstorm on the Loess Plateau, China. *International Journal of Sediment Research*, 35(4), 408-416. [link]

27. Wang, H., Jiao, J., Tang, B., **Chen, Y.**, Bai, L., Wang, N., et al. (2019). Characteristics of Rill Erosion and Its Influencing Factors in Slope Farmland After "7 | 26" Rainstorm in Zizhou County, Shaanxi Province. *Transactions of the Chinese Society of Agricultural Engineering*, 35(11), 122-130. (EI-indexed, in Chinese with English abstract) [link]

28. Zhang, Y., Jiao, J., **Chen, Y.**, & Tang, B. (2019a). Channel Morphology and Sediment Connectivity of Fangta Watershed in Ansai County of Northern Shaanxi Province. *Research of Soil and Water Conservation*, 26(03), 11-15. (In Chinese with English abstract) [link]

29. Zhang, Y., Jiao, J., Tang, B., **Chen, Y.**, Wang, N., Bai, L., et al. (2019b). Channel Sediment Connectivity and Influence Factors in Small Watersheds under Extreme Rainstorm - A Case Study at Zizhou County, Shaanxi Province. *Bulletin of Soil and Water Conservation*, 39(1), 302-309. (In Chinese with English abstract) [link]

30. Zhao, C., Jiao, J., Shui, J., Wang, H., Chen, T., **Chen, Y.**, et al. (2019). Comparative Analysis on Morphological Characteristics of Erosion Gullies Measured by an Unmanned Aerial Vehicle and Traditional Ground Survey in South Central Tibet. *Bulletin of Soil and Water Conservation*, 39(5), 120-127. (In Chinese with English abstract) [link]

31. Wei, Y., He, Z., Jiao, J., Li, Y., **Chen, Y.**, & Zhao, H. (2018). Variation in the Sediment Deposition Behind Check-Dams Under Different Soil Erosion Conditions on the Loess Plateau, China. *Earth Surface Processes and Landforms*, 43(9), 1899-1912. [link]

32. Hu, S., Jiao, J., García-Fayos, P., Kou, M., **Chen, Y.**, & Wang, W. (2018). Telling a Different Story: Plant Recolonization After Landslides Under a Semi-Arid Climate. *Plant and Soil*, 426(1-2), 163-178. [link]

33. Wang, N., **Chen, Y.**, Bai, L., Wang, H., & Jiao, J. (2017). Investigation on Soil Erosion in Small Watersheds Under "7 | 26" Extreme Rainstorm in Zizhou County, Northern Shaanxi Province. *Bulletin of Soil and Water Conservation*, 37(4), 338-344. (In Chinese with English abstract) [link]

34. Wu, D., Jiao, J., Yu, W., Xu, H., & **Chen, Y**. (2017). Response Characteristics of Locust Growth, Understory Vegetation and Soil Moisture on Planting Density in Northern Shaanxi. *Acta Botanica Boreali-Occidentalia Sinica*, 37(2), 346-355. (In Chinese with English abstract) [link]

BOOK CHAPTER

Jiao, J., Wang, N., Liang, Y., **Chen, Y.**, Tang, B., Bai, L., *et al*. Effectiveness of Soil and Water Conservation Measures. In: Comprehensive Investigation Report on Soil and Water Conservation Effectiveness Against the Erosion and Flooding by the "7 | 26" Exceptional Rainstorm in the Chinese Loess Plateau. Liu, B., Yao, W., Liu, G., An, S., Han, J., Cao, W., *et al*. First Edition. *Science Press*, Beijing, China. 2020. Chapter 10: 188-215. [link]

DATASETS AND PYTHON TOOLS

Chen, Y., Cohen, S., Dhital, S., Baruah, A., & Devi, D. (2026). Improved River Slope Datasets for the United States Hydrofabrics (Versions v1 - v5) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.21632749 (total downloads: 915)

Chen, Y., Baruah, A., Devi, D., & Cohen, S. (2026). Improved River Slope Datasets for the United States Hydrofabrics, HydroShare, https://doi.org/10.4211/hs.1532f4cb360244f9a6ba772ebd428180 (views: 2154; downloads: 149)

Chen, Y., Devi, D., Cohen, S., Dhital, S., Baruah, A., Frasson, R. P. M., & Pavelsky, T. M. (2026). Global SWOT River Water Surface Slope Observations (2023-2026), Reach-Level Temporal Variability Statistics, and Gauge-Pair Slope Code (Version 1) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.23023286 (downloads: 98)

Chen, Y., Cohen, S., Dhital, S., Devi, D., & Baruah, A. (2025). Global Monthly Reach-Level Water Surface Slopes from SWOT [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.17926443 (downloads: 270)

Chen, Y., Dhital, S., Baruah, A., Devi, D., Munasinghe, D., & Cohen, S. Framework for Spatially Joining Two Hydrographic Datasets through the Hydrofabric Flowlines - riverjoin. GitHub Repo. June 17, 2026. https://github.com/sdmlua/riverjoin_py

Chen, Y., Dhital, S., Baruah, A., Devi, D., Munasinghe, D., & Cohen, S. Framework for Spatially Joining Two Hydrographic Datasets through the Hydrofabric Flowlines - riverjoin 0.1.11. Python package. June 17, 2026. https://pypi.org/project/riverjoin/ (total downloads: 2,527, per pepy.tech)

CONFERENCE PRESENTATIONS

1. **Chen, Y.**, Baruah, A., Devi, D., Phillips, C., Lane, B., & Cohen, S. (2026, September). Sensitivity-Informed Synthetic Rating Curve (SRC) Calibration to Improve OWP HAND-FIM Performance. In: *NOAA CIROH Science Meeting*, Tuscaloosa, AL. (Oral)

2. **Chen, Y.**, Cohen, S., Dhital, S., Baruah, A., Devi, D., & Munasinghe, D. (2026, August). RiverJoin: A Framework for Spatially Joining River Flowlines Across Heterogeneous Hydrographic Datasets to Enable Bidirectional Attribute Transfer. In: *Global Flood Partnership Annual Meeting*, Provo, UT. (Oral)

3. **Chen, Y.**, Dhital, S., Cohen, S., Baruah, A., Devi, D., & Zarrabi, R. (2026, May). Temporal Variability in Global River Slopes from the Surface Water and Ocean Topography (SWOT) Satellite. In: *NOAA CIROH Developers Conference*, Salt Lake City, UT. (Poster)

4. **Chen, Y.**, Dhital, S., Cohen, S., Baruah, A., & Devi, D. (2025). Global-Scale Analysis of River Slope Monthly Variation Based on the SWOT River Vector Data Product. In: *American Geophysical Union (AGU) Annual Meeting*, New Orleans, LA. (Poster)

5. **Chen, Y.**, Baruah, A., Devi, D., & Cohen, S. (2025). Improved River Slope Datasets for the United States Operational Flood Inundation Mapping Hydrofabric and Next Generation Water Resources Modeling Hydrofabric. In: *NOAA CIROH Developers Conference*, Burlington, VT. (Poster)

6. **Chen, Y.**, Baruah, A., Devi, D., & Cohen, S. (2024). Updating the US Hydrofabric Slope Parameter Using a Remote Sensing Dataset: Impact on Flood Inundation Mapping. *AGU2024 Annual Meeting*, Washington, DC. (Poster) [link]

7. **Chen, Y.**, De Geeter, S., Poesen, J. W., Matthews, F., Campforts, B., Borrelli, P., ... & Vanmaercke, M. (2024, December). Data-Driven Prediction of Gully Densities at the Global Scale. *AGU2024 Annual Meeting*, Washington, DC (Vol. 2024, No. 63, pp. GC31R-0063). (Poster)

8. **Chen, Y.**, De Geeter, S., Poesen, J., Campforts, B., Borrelli, P., Panagos, P., & Vanmaercke, M. (2023). Data-Driven Prediction of Gully Densities at the Global Scale. *Young Soil Scientists Day of Soil Science Society of Belgium*, Brussels, Belgium. (Oral)

9. **Chen, Y.**, De Geeter, S., Poesen, J., Campforts, B., Borrelli, P., Panagos, P., & Vanmaercke, M. (2023). Data-Driven Prediction of Gully Densities at the Global Scale. *The 9th International Symposium on Gully Erosion*, Chengdu, China. (Oral)
10. **Chen, Y.**, Jiao, J., Yan, X., & Li, J. (2023). Characteristics of Morphology and Density of Gullies Along a Rainfall Gradient on the Chinese Loess Plateau. *Annual Meeting of Soil Erosion and Soil and Water Conservation Professional Committee of China Soil Society*, Xichang, China. (Oral)
11. **Chen, Y.**, Vanmaercke, M., Jiao, J., Bai, L., Tang, B., Wang, N., et al. (2021). Quantifying the Importance of Different Erosion Processes and Collapsed Soil and Water Conservation Measures on Catchment Sediment Due to an Extreme Rainstorm: A Field-Based Case Study in the Chinese Loess Plateau. *International Forum on Land Degradation, Soil Conservation and Sustainable Development*, Dalian, China. (Poster)
12. **Chen, Y.**, Jiao, J., Vanmaercke, M., Yan, X., & Li, J. (2020). Spatial Variation of Gully Density and Morphology and Their Controlling Factors at a Regional Scale: A Case Study for the Chinese Loess Plateau. In *EGU General Assembly Conference Abstracts*, Vienna, Austria, and online, p. 147. (Poster) [link]
13. **Chen, Y.**, Jiao, J., Bai, L., Tang, B., Wang, N., Zhang, Y., & Wang, H. (2018). Soil Erosion Hotspots in Small Watershed Under Extreme Rainstorm on the Loess Plateau, China. *The Second International Young Scientists Forum on Soil and Water Conservation and ICCE symposium 2018 "Climate Change Impacts on Sediment Dynamics: Measurement, Modelling and Management"*, Moscow, Russia. (Oral)
14. Vanmaercke, M., **Chen, Y.**, De Geeter, S., Yihunie, Y., Dujardin, E., Mawe, G. I., Landu, E. L., & Poesen, J. (2025). Gully Erosion Might Become a Larger Problem than Hitherto Anticipated: Insights from Fieldwork and Recent Modelling Advancements. In *EGU General Assembly Conference Abstracts*, Vienna, Austria, EGU25-9577. (Invited oral) [link]
15. Liu, H., Wang, C., **Chen, Y.**, Ma, L., Zhang, C., Long, Y., & Yang, Q. (2025). Random Forest-Based Prediction of Gully Density on Farmland in the Songnen Typical Black Soil Region of the Northeast, China. In *EGU General Assembly Conference Abstracts*, Vienna, Austria, EGU25-9463. (Oral) [link]
16. Vanmaercke, M., **Chen, Y.**, De Geeter, S., Poesen, J., Campforts, B., Borrelli, P., et al. (2022). Data-Driven Prediction of Gully Densities and Erosion Risk at the Global Scale. In *EGU General Assembly Conference Abstracts*, Vienna, Austria, EGU22-2921. (Oral) [link]
17. Vanmaercke, M., **Chen, Y.**, De Geeter, S., Poesen, J., & Campforts, B. (2020). A First Data-Driven Gully Head Density Map of the World. In *EGU General Assembly Conference Abstracts*, Vienna, Austria, and online, p. 9285. (Oral) [link]

TEACHING AND (CO-)SUPERVISING

Fall 2026 - present	GIS Programming - Teaching Assistant, The University of Alabama; teaches class sessions of the course
2026	Co-led graduate students assessing the impact of dynamic river slope on flood inundation mapping in the CUAHSI Water Prediction Innovators Summer Institute at the University of Alabama
2025 - present	Supervising 5 undergraduate students at Northwest University (China) to identify the effect of hillslope gullies on hydrological connectivity and downstream soil erosion
2025	GIS Programming - Guest lecture, The University of Alabama
2023 - 2026	Co-supervised 1 graduate student at Northwest University (China) on her Master Thesis on gully occurrence and morphology modeling in the Northeastern Black Soil region of China
2019 - 2024	Co-supervised 1 graduate student at the Institute of Soil and Water Conservation, Chinese Academy of Sciences on his PhD Dissertation on gully erosion modeling using machine learning and gully systematic classification in the Tibetan Plateau. Now an Assistant Professor at Shanxi Agricultural University, China

FIELD RESEARCH EXPERIENCE

Nearly a decade of fieldwork on the Chinese Loess Plateau.

Gully monitoring	Monitored gully development with on-site erosion pins, repeated RTK GNSS surveys, and multiple UAV photogrammetry (Structure-from-Motion) campaigns
Post-event surveys	Catchment-scale erosion surveys after extreme rainstorms, quantifying rill erosion, gully erosion, shallow landslides, and the failure of soil and water conservation structures (bench terraces and check dams)
Runoff plots	Established runoff plots along a vegetation-restoration chronosequence and installed rain gauges; collected runoff samples and precipitation records

Vegetation and soils	Surveyed vegetation species composition, community structure, cover, and plant growth characteristics; collected soil and vegetation samples
Landslides	Participated in field investigations of vegetation recovery on landslides
Connectivity	Assisted studies of hillslope runoff and hydrological connectivity using field observation of flow pathways and rare earth element (REE) tracers
2026.03	Led five undergraduates through a UAV photogrammetry field campaign at Ansai, Loess Plateau (NSFC Young Scientist project)

PROFESSIONAL SERVICE

Peer review. 30 reviews for 11 journals, 2021-2025: Catena (10), Sustainability (6), Advances in Space Research (4), Geomorphology (2), Soil & Tillage Research (2), and one each for Anthropocene, Earth Surface Processes and Landforms, Geoderma Regional, Geosciences, Hydrology and International Journal of Sediment Research.
Conference convening. Co-convener, Session SSS2.3, "Gully and Rill Erosion: Recent Advancements and Novel Approaches," EGU General Assembly 2026.
Grant review. Proposal reviewer, National Natural Science Foundation of China (NSFC), since 2025.
Institutional review. Invited by the Director of NOAA CIROH to present to the Institute's external review panel at its 5 Year Science Review, March 2026.

MEDIA COVERAGE

Cooperative Institute for Research to Operations in Hydrology (CIROH) News: "New Satellite-Derived River Slope Database Helps Improve U.S. Flood Forecasts" (https://awi.ua.edu/2025/09/26/new-satellite-derived-river-slope-database-helps-improves-u-s-flood-forecasts/), featuring my research on improving river slope representation to increase the accuracy of NOAA operational flood inundation mapping.

(CIROH, a partnership between NOAA and The University of Alabama, is a national consortium committed to advancing water prediction - the forecasting of streamflow entering water systems, extreme events such as floods and droughts, and water quality - and building community resilience to water-related challenges.)

BBC News Brasil: "The vast ravines swallowing whole neighborhoods around the world" (https://www.bbc.co.uk/news/resources/idt-61d57e25-d458-4785-8a37-3ef119a504c7), highlighting results from our global gully erosion research.

EU Soil Observatory Newsletter by European Commission's Joint Research Centre: "Global patterns of gully occurrence" (https://ec.europa.eu/newsroom/jrc/newsletter-archives/74237), featuring my global gully occurrence modeling work as a research highlight; the resulting global gully maps are archived in the European Soil Data Centre (ESDAC, https://esdac.jrc.ec.europa.eu/content/global-patterns-gully-occurrence).

ScienceNet: "The hazards and considerations of soil erosion in small watersheds caused by the 'July 26' heavy rainstorm in Zizhou, northern Shaanxi, China" (https://news.sciencenet.cn/sbhtmlnews/2017/9/327683.shtm), featuring our research findings on severe soil erosion induced by heavy rainstorms and efforts to control it.

(ScienceNet is a science news website supervised by the Chinese Academy of Sciences, the Chinese Academy of Engineering and the National Natural Science Foundation of China, and hosted by Science Times.)

SCIENCE COMMUNICATION AND OUTREACH

Dhital, S., Devi, D., Baruah, A., Munasinghe, D., Nikrou, P., Tian, D., **Chen, Y.**, & Cohen, S. (2026, August 20). *CIROH Flood Inundation Mapping (FIM) Community Ecosystem: A Large-Scale Operational Flood Modeling-to-Evaluation Framework*. CIROH Hub. https://hub.ciroh.org/blog/fim-community-ecosystem/